

Module 6



Assignment: Exploratory analysis and presenting insights

2Market Data Analysis Project Report

Problem statement and approach:

2Market is a global supermarket selling products both in-store and online, serving a diverse customer base across multiple countries. The company required a deeper understanding of its customer purchasing behaviour based on demographic groups to launch a future marketing campaign strategy and better data-driven decision making [1].

2Market wants to analyse data and understand relationships between demographics such as age, income, marital status, education level, and country influence spending patterns, and which products contribute most to the total revenue.

Additionally, 2Market aims to evaluate the effectiveness of its advertising channels, and conversion performance across Twitter, Bulkmail, Instagram, Facebook, and brochure campaigns.

By exploring the datasets, insights will be generated that can help with targeted marketing decisions, budget allocation and in identifying high-value customer segments for future growth.

Analytical Approach:

This project follows the IDEAL [2] problem-solving model, which shows progression from data understanding to analysis and evaluation.

I – Identify the problem and opportunities.

2Market needs understanding of who their customers are, which advertising channels are most effective, and purchase influence according to demographics. These insights support future marketing campaign strategies.

D - Define goals

The problem is presented with two datasets makerting_data and ad-data files. These have information including age, income, marital status, education level, product categories (e.g., AmtLiq, AmtVege, NonVeg, Chocholates) and variables for Twitter, Insta, Facebook, Bulkmail, and Brochure ads. Both datasets are joined using SQL (left join) on customer ID to create a unified dataset.

E - Explore Possible Solutions and Strategies

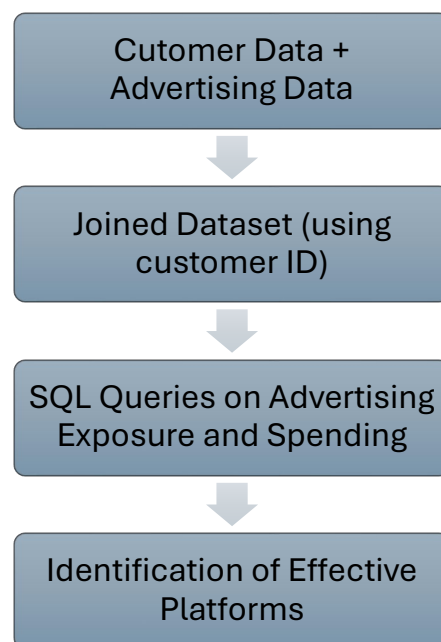
2Market customer dataset is cleaned in Microsoft Excel and checked for missing values and duplicate customer IDs. Income values are converted from text to numeric by removing currency symbols \$ and commas. Age is calculated using Year_Birth (2014 used as reference year), the average age is 44 years, with most customers falling in middle-age range and married customers appear as largest demographic cluster.

A scatter plot visualisation (see Appendix A) shows no evidence of strong age-income linear relationship, confirming age is not a strong predictor of earnings.

SQL analysis follows these steps:

- Tables created for marketing_data and ad_data.
- Null IDs cleaned.
- Left join used for all records.
- Aggregate found by country and marital status.
- Total spending by product groups calculated.

The following flowchart represents the data-processing workflow used in this project. It outlines how customer and advertising datasets were joined via customer ID, analysed through SQL exposure and spending aggregations, and used to identify the advertising platforms generating the strongest engagement and conversions.



Visual diagram of SQL flow

These queries identified which platforms are effectively used by demographic groups and how do they align with customer spending.

Tableau explored visual patterns for demographic distribution, spending trends, and platform effectiveness which confirm that spending is heavily skewed towards alcoholic and meat products. Highly educated customers show more average spend.

Advertising analysis confirms that digital ad channels such as Twitter, Instagram performed best.

A - Anticipate Outcomes

SQL and Tableau insights are combined to summarise actionable findings. Visuals such as advertising channel “Effectiveness per Marital Status” and “Effectiveness per Country” provides straightforward evidence of channel performance patterns.

L - Look back and Learn

Evaluation confirms limitations such as:

- no gender variable
- no confirm conversions of advertising exposure
- spending values lack time-series detail.

Data Analysis tools such as Excel, SQL and Tableau proved effective in cleaning, aggregating and for identifying patterns.

Improvement could include using time bound purchase data, running a channel test, and adding more demographic attributes for targeting accuracy.

Dashboard design and development:

A Tableau dashboard is designed to present insights from the data for 2Market about:

- Customer demographics
- Advertising Effectiveness
- Customer spending Behaviour

It pulls key findings together from Excel, SQL, and Tableau analysis into a set of visuals which support facts and informed decision making.

There are three KPI cards that summarise business performance as:

- Total revenue - \$1,343,277
- Total Customers – 2,2,12
- Average age – 44 years

KPIs added to the top of the dashboard, give a quick overview of the scale and customer characteristics to stakeholders.

Datasets are visualised based on the type of insights:

- Histograms for Customer Age Distribution because it shows concentration of customers in 40-60 age range, which is middle-aged customer base.
- Bar Charts for Average Spend by Age Group, consolidating spending increases with age peaking in customers 60+.
- A map for Total Sales by Country for geographic interpretation showing Spain has most customers.

Advertising Channel Effectiveness dashboard compares digital and print channels which confirms performance gaps between Twitter, Bulk mail, Instagram, Facebook, and Brochures.

The dashboard includes interactive filters for age groups, country, marital status, and education. These filters allow isolating specific demographic segments and make targeted comparisons.

For example, filters can help assess whether certain age groups respond more to a particular advertising channel or if spending differs by country.

The dashboard is simple, readable, and accessible. Consistent chart layouts, labelling, and spacing ensures readability and inclusivity for colour-blind-friendly palette.

Patterns, trends, and insight

The analysis shows various behaviour patterns across 2Market's customers. Demographically, majority of customers are aged between 40 and 60, with an average age of 44, confirming middle-age as predominant profile.

Spending patterns strongly correlate with age: customers aged 60+ spend the most on average, while customers under 30 spend less.

Spending behaviour also changes with product categories. Alcoholic beverages and non-veg products generate highest revenue which is far high than confectionary or vegetables.

Education level depicts a positive relationship with spending. PhD, Master's, and Graduation-level customers spend more. Higher-educated segments highlight greater lifetime value.

Geographically, Spain (SP) has most customers base, but ME and SA record higher average spend per customer which accentuates potential high-value regional markets for targeted promotions.

Advertising analysis show that digital channels outperform print. Twitter, Bulkmail, Instagram, and Facebook generate more successful conversions than brochure advertising. Approximately 20% of customers respond at least one ad but responses differ by country and education level.

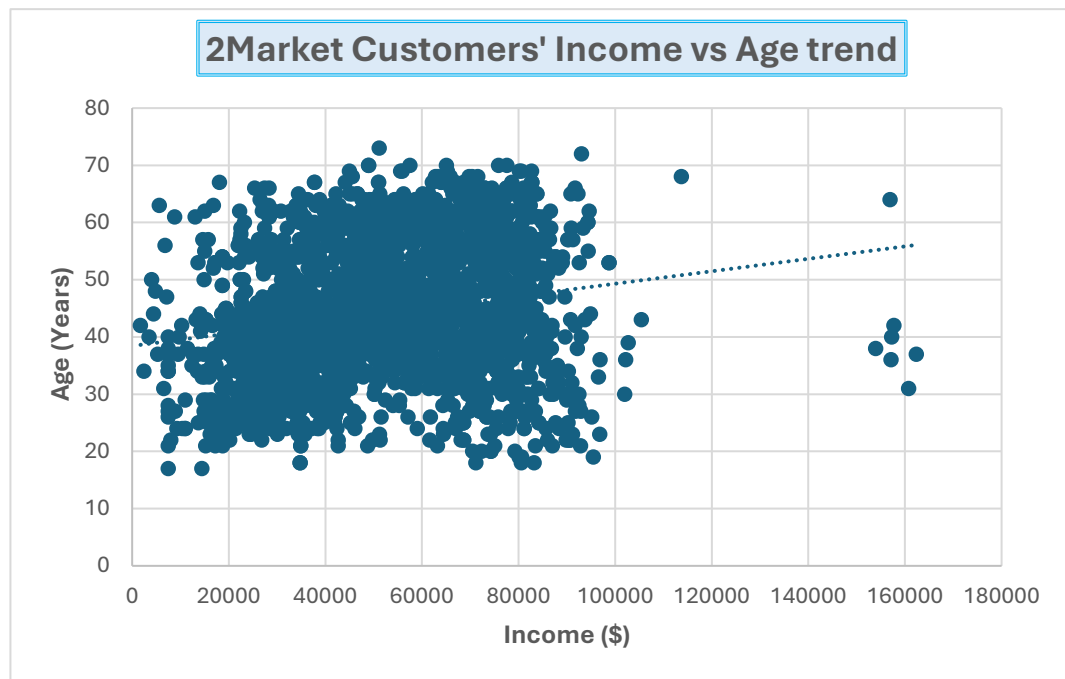
These insights help with advertisement budgets allocation by prioritising digital channels and planning campaigns to high-value demographic segments.

References

1. Ticong, L. (2024) *What is Data-Driven Decision Making? 6 Key Steps (+ Examples)*. Available at : [Data-Driven Decision-Making: 6 Key Steps \(+ Examples\)](#) (Accessed: 1 April 2026)
2. Hanson, C. (2021) *Why the IDEAL Problem-Solving Method Breaks Down for ND Brains (And How to Fix Each Step)*, Life Skills Advocate. Available at: [\(5 Steps Of The IDEAL Problem-Solving Method \(and Why Each One Is Hard For ND Brains\) | Life Skills Advocate](#) (Accessed: 2 April 2026)
3. Cotgreave, A. (2024) *My Tableau Story: Andy Cotgreave*. Tableau Public. Available at: [My Tableau Story: Andy Cotgreave](#) (Accessed: 2 April 2026)
4. Royal Statistical Society (2023) *Best Practices for Data Visualisation*. Available at: <https://royal-statistical-society.github.io/datavisguide/> (Accessed: 4 April 2026).
5. Kothari, H.K. et al. (2025) 'Impact of social media advertising on consumer behavior: role of credibility, perceived authenticity, and sustainability', *Frontiers in Communication*, 10. Available at: <https://www.frontiersin.org/journals/communication/articles/10.3389/fcomm.2025.1595796/full> (Accessed: 4 April 2026).
6. W3 schools (2026) *SQL Joins. The SQL JOIN Clause*. Available at : [SQL Joins](#) (Accessed: 5 A
7. AI is used in this project to troubleshoot syntax errors and to gain ideas how to clean data such as to change date formats.

Appendix

(A)



SQL Script

```
/* Create Table for marketing_data
```

```
*/
```

```
CREATE TABLE marketing_data (
```

```
    ID INT,
```

```
    Year_Birth INT,
```

```
    Education VARCHAR(50),
```

```
    Marital_Status VARCHAR(50),
```

```
    Income INT,
```

```
    Kidhome INT,
```

```
    Teenhome INT,
```

```
    Dt_Customer DATE,
```

```
    Recency INT,
```

```
    AmtLiq INT,
```



```
AmtVege INT,  
AmtNonVeg INT,  
AmtPes INT,  
AmtChocolates INT,  
AmtComm INT,  
NumDeals INT,  
NumWebBuy INT,  
NumWalkinPur INT,  
NumVisits INT,  
Response INT,  
Complain INT,  
Country VARCHAR(20),  
Count_success INT,  
Age INT  
);
```

```
ALTER TABLE marketing_data  
ADD PRIMARY KEY (ID);
```

```
DELETE FROM marketing_data WHERE ID IS NULL;  
ALTER TABLE marketing_data ADD PRIMARY KEY (ID);
```

```
SELECT COUNT(*) FROM marketing_data;
```

```
SELECT * FROM marketing_data LIMIT 10;
```

```
/* Create table for ad_data*/
```

```
DROP TABLE IF EXISTS ad_data;
```

```
CREATE TABLE ad_data (  
    ID INT,  
    Bulkmail_ad INT,
```

```

    Twitter_ad INT,
    Instagram_ad INT,
    Facebook_ad INT,
    Brochure_ad INT
);

SELECT * FROM ad_data LIMIT 10;

/* Join the TWO Tables*/

DROP TABLE IF EXISTS marketing_full;

CREATE TABLE marketing_full AS
SELECT
    m.*,
    a.Bulkmail_ad,
    a.Twitter_ad,
    a.Instagram_ad,
    a.Facebook_ad,
    a.Brochure_ad
FROM marketing_data m
LEFT JOIN ad_data a
ON m.ID = a.ID;

SELECT COUNT(*) FROM marketing_full;

/* Most effective platform per country
*/

SELECT
    Country,
    'Twitter' AS Platform,
    SUM(Twitter_ad) AS Total_Exposure

```

```
FROM marketing_full
```

```
GROUP BY Country
```

```
UNION ALL
```

```
SELECT
```

```
Country,
```

```
'Instagram',
```

```
SUM(Instagram_ad)
```

```
FROM marketing_full
```

```
GROUP BY Country
```

```
UNION ALL
```

```
SELECT
```

```
Country,
```

```
'Facebook',
```

```
SUM(Facebook_ad)
```

```
FROM marketing_full
```

```
GROUP BY Country
```

```
ORDER BY Country, Total_Exposure DESC;
```

```
/* Most effective platform per marital status */
```

```
SELECT
```

```
Marital_Status,
```

```
'Twitter' AS Platform,
```

```
SUM(Twitter_ad) AS Total_Exposure
```

```
FROM marketing_full
```

```
GROUP BY Marital_Status
```

```
UNION ALL
```

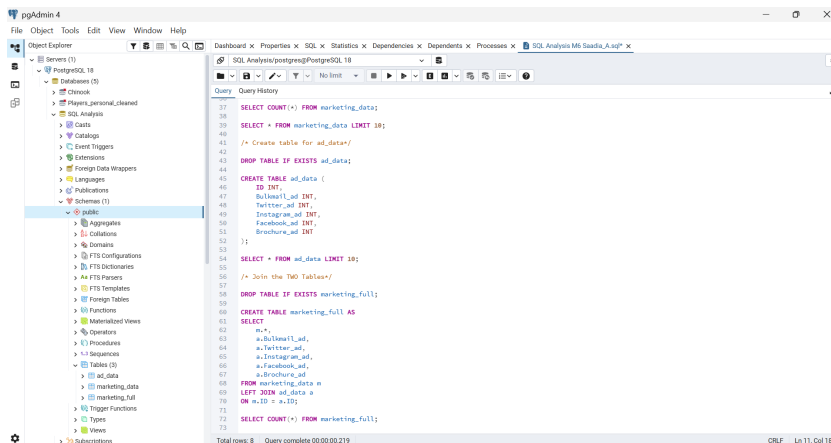
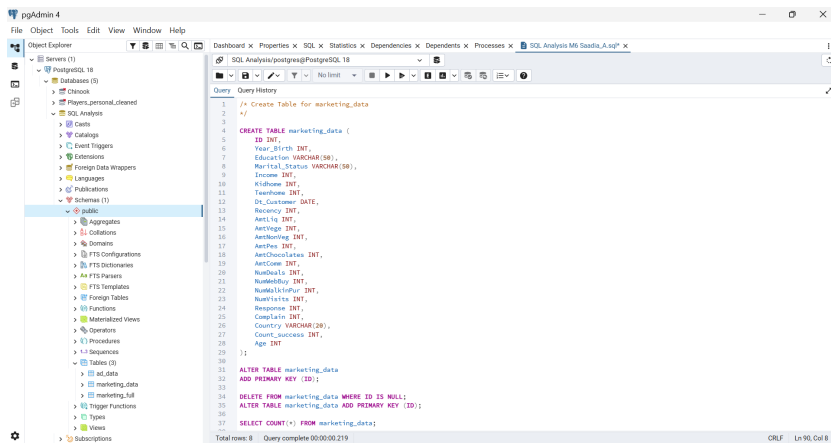
```
SELECT
    Marital_Status,
    'Instagram',
    SUM(Instagram_ad)
FROM marketing_full
GROUP BY Marital_Status
```

```
UNION ALL
```

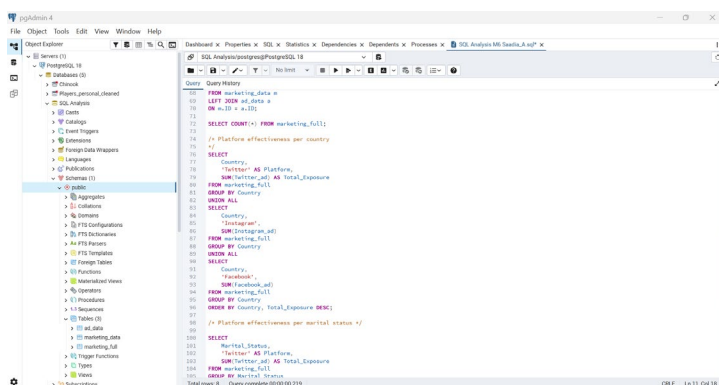
```
SELECT
    Marital_Status,
    'Facebook',
    SUM(Facebook_ad)
FROM marketing_full
GROUP BY Marital_Status
ORDER BY Marital_Status, Total_Exposure DESC;
```

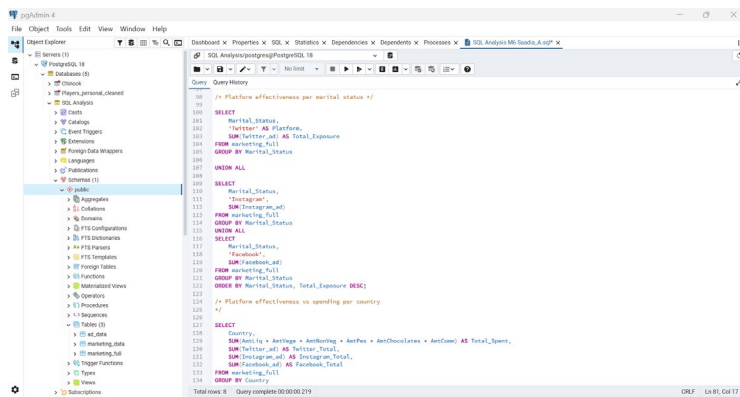
```
/* Relationship between ad platform and total spending per country
*/
```

```
SELECT
    Country,
    SUM(AmtLiq + AmtVege + AmtNonVeg + AmtPes + AmtChocolates + AmtComm) AS Total_Spent,
    SUM(Twitter_ad) AS Twitter_Total,
    SUM(Instagram_ad) AS Instagram_Total,
    SUM(Facebook_ad) AS Facebook_Total
FROM marketing_full
GROUP BY Country
ORDER BY Total_Spent DESC;
```



	country character varying (20)	platform text	total_exposure bigint
1	AUS	Instagram	12
2	AUS	Facebook	7
3	AUS	Twitter	6
4	CA	Twitter	24
5	CA	Instagram	21
6	CA	Facebook	18
7	GER	Twitter	11
8	GER	Instagram	8
9	GER	Facebook	7
10	IND	Twitter	10
11	IND	Facebook	7
12	IND	Instagram	6
13	ME	Twitter	0
14	ME	Instagram	0
15	ME	Facebook	0
16	SA	Instagram	21
17	SA	Facebook	20
18	SA	Twitter	20
19	SP	Instagram	88
20	SP	Twitter	87
21	SP	Facebook	76
22	US	Facebook	7
23	US	Twitter	6
24	US	Instagram	5





	marital_status character varying (50)	platform text	total_exposure bigint
1	Divorced	Twitter	18
2	Divorced	Instagram	13
3	Divorced	Facebook	12
4	Married	Instagram	66
5	Married	Facebook	62
6	Married	Twitter	62
7	Single	Twitter	32
8	Single	Instagram	31
9	Single	Facebook	30
10	Together	Instagram	43
11	Together	Twitter	42
12	Together	Facebook	32
13	Unknown	Facebook	1
14	Unknown	Instagram	1
15	Unknown	Twitter	0
16	Widow	Twitter	10
17	Widow	Instagram	7
18	Widow	Facebook	5

